

BERGAMO ORIO AL SERIO, ItalyWMO: **160760**Lat: **45.669N**Lon: **9.700E**Elev: **778**StdP: **14.29**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.8	27.5	9.0	9.0	33.1	13.9	11.5	34.1	17.8	52.9	14.4	49.3	3.9	10	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	91.1	73.2	88.2	71.6	86.1	70.6	75.1	87.5	73.3	85.9	71.8	83.4	7.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.0	117.4	83.6	69.0	109.8	81.4	67.7	104.8	79.6	39.0	88.0	37.4	85.8	35.9	83.4	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.7	12.0	10.0	20.9	94.0	3.8	2.6	18.2	95.9	15.9	97.5	13.8	99.0	11.0	100.9		
(5)			18.5	77.0	3.5	2.2	16.0	78.6	13.9	79.9	11.9	81.2	9.3	82.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	39.0	41.4	49.1	55.9	64.1	71.7	75.6	74.9	66.8	58.1	47.9	40.1	
	DBStd	14.10	4.93	5.43	5.98	5.73	5.82	5.94	4.61	5.03	5.08	5.38	5.76	5.32	
	HDD50	1105	341	243	87	14	0	0	0	0	0	6	104	310	
	HDD65	3879	805	661	494	276	87	12	0	1	38	220	512	773	
	CDD50	3710	1	2	58	191	438	651	794	771	503	258	42	1	
	CDD65	1011	0	0	0	3	60	213	330	307	91	7	0	0	
	CDH74	8575	0	0	4	25	442	1860	3068	2660	495	21	0	0	
	CDH80	2845	0	0	0	2	88	612	1111	955	76	1	0	0	
Wind	WSAvg	4.5	3.6	4.2	5.1	5.1	5.2	5.3	5.3	5.0	4.4	3.9	3.5	3.3	
Precipitation	PrecAvg	37.7	2.0	1.9	2.0	3.5	4.2	3.6	2.7	3.5	4.1	4.3	4.4	2.2	
	PrecMax	67.3	9.8	7.3	6.6	7.3	8.7	11.2	7.8	7.8	14.5	10.3	13.7	6.3	
	PrecMin	23.7	0.0	0.0	0.1	0.6	0.9	0.4	0.4	0.2	0.7	0.7	0.2	0.0	
	PrecStd	10.7	2.0	2.0	1.5	1.6	2.2	2.4	2.0	2.0	3.2	2.5	3.5	1.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	62.5	71.8	77.1	86.4	92.5	93.5	94.9	85.2	76.7	64.7	56.7	
		MCWB	46.4	49.9	55.1	61.0	68.4	74.5	73.2	75.0	69.0	64.4	56.0	47.9	
	2%	DB	52.1	56.8	66.4	73.3	82.2	89.4	90.5	91.1	82.3	72.5	60.9	53.2	
		MCWB	44.4	47.3	52.5	58.7	65.9	71.8	73.2	73.1	67.4	61.9	53.8	46.3	
	5%	DB	49.6	53.5	62.9	70.1	78.8	86.2	88.1	88.0	79.4	69.7	58.4	50.1	
		MCWB	43.2	44.8	51.4	57.0	63.4	70.5	71.8	71.5	65.9	60.7	53.2	45.2	
	10%	DB	46.6	50.2	59.7	67.0	75.5	83.8	86.0	85.6	76.8	66.6	55.7	47.9	
		MCWB	42.2	43.4	49.7	55.4	62.2	68.8	70.8	70.6	64.0	58.9	51.8	43.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	50.4	56.4	62.1	70.1	76.7	77.3	77.4	71.1	65.5	57.8	51.6
			MCDB	53.3	58.9	67.1	74.3	83.5	90.3	88.6	91.9	82.0	72.7	62.0	54.0
2%		WB	46.6	48.1	54.0	59.8	67.0	73.5	74.9	75.0	69.1	63.8	55.7	48.6	
		MCDB	49.3	54.3	63.5	70.7	79.4	86.4	86.8	87.8	78.8	69.8	59.1	50.6	
5%		WB	44.6	46.2	52.3	58.1	64.8	71.4	73.3	73.2	67.4	62.2	54.2	45.9	
		MCDB	46.9	50.5	60.4	68.1	75.9	83.5	85.7	85.6	76.4	67.6	57.4	48.1	
10%		WB	42.6	44.6	50.7	56.4	63.0	69.8	71.9	71.3	65.7	60.4	52.7	44.2	
		MCDB	45.3	48.1	57.7	65.1	73.5	81.8	83.4	82.3	74.2	65.6	55.7	46.5	
Mean Daily Temperature Range		5% DB	MDBR	12.6	14.1	16.2	16.2	17.1	17.7	18.1	17.4	15.8	13.2	11.4	12.0
			MCDBR	17.1	19.8	21.6	20.8	21.0	19.9	19.3	19.5	18.5	16.9	14.3	15.7
	MCWBR		11.1	12.6	12.2	10.6	9.7	8.9	8.6	8.4	7.9	8.7	9.0	10.5	
	5% WB	MCDBR	13.2	16.2	18.5	18.6	19.0	19.0	18.1	18.6	16.4	14.1	10.7	11.5	
		MCWBR	9.5	11.0	11.0	9.8	9.3	9.0	8.9	9.1	7.7	7.8	7.3	8.7	
Clear-Sky Solar Irradiance	taub	0.329	0.361	0.409	0.452	0.452	0.475	0.456	0.445	0.438	0.419	0.362	0.325		
	taud	2.383	2.286	2.181	2.097	2.151	2.130	2.191	2.231	2.218	2.232	2.357	2.420		
	Ebn at Noon	243	254	257	256	260	254	258	255	245	229	226	231		
	Edh at Noon	24	31	40	48	47	48	44	41	38	33	24	21		
All-Sky Solar Radiation	RadAvg	466	727	1144	1427	1724	1989	2023	1721	1294	779	464	390		
	RadStd	66	102	149	170	148	122	128	114	94	115	78	54		